Supplemental Material for "The UZH protocol: Separating and reducing Gaussian-basis and pseudopotential errors in CP2K"

Hossein Mirhosseini,[1,2] Tiziano M. A. Müller,[3] Matthias Krack,[4] Thomas D. Kühne,[1,2,5,*] and Jürg Hutter[3]

[1]*Center for Advanced Systems Understanding (CASUS), Conrad-Schiedt-Straße 20, 02826 Görlitz, Germany*

[2]*Helmholtz-Zentrum Dresden-Rossendorf, Bautzner Landstraße 400, 01328 Dresden, Germany*

[3]*Department of Chemistry, University of Zurich, CH-8057 Zurich, Switzerland*

[4]*Paul Scherrer Institute, 5232 Villigen PSI, Switzerland*

[5]*Institute of Artificial Intelligence, Technische Universität Dresden, Helmholtzstraße 10, 01069 Dresden, Germany*

I. SOURCE MATERIAL AND IMPLEMENTATION DETAILS

The main text summarizes the protocol, while this Supplemental Material records additional details that are useful for reproduction and for future development of the UZH settings. The organization follows the original molecularly optimized (MOLOPT), Goedecker–Teter–Hutter (GTH), and Hartwigsen–Goedecker–Hutter (HGH) literature [1–3], the revised MOLOPT benchmarking work [4], the CP2K implementation papers [5, 6], and the Automated Interactive Infrastructure and Database for Computational Science (AiiDA) verification protocol and its Supplemental Information [7, 8].

A. CP2K ATOM and basis optimization

The updated CP2K ATOM implementation developed for the present UZH work is exposed through a dedicated input section for atomic calculations. The relevant run types are energy calculations, basis optimization, and pseudopotential optimization. The ATOM method section supports restricted and unrestricted Kohn-Sham and Hartree-Fock atomic calculations and scalar-relativistic Hamiltonians including zeroth-order regular approximation (ZORA)-type and Douglas–Kroll–Hess variants. The relativistic section allows the order of the Douglas–Kroll–Hess transformation, the transformation strategy, and the ZORA model-potential variant to be selected. The potential section supports all-electron, GTH, unified pseudopotential format, separable Gaussian pseudopotential, and effective core potential inputs. For the GTH format, the input layer exposes a CP2K-standard GTH potential block, and the print layer can write the optimized potential in CP2K format and in a form compatible with the unified pseudopotential format.

The CP2K basis optimizer works from template, work, and output basis files. It supports simultaneous fitting of related basis levels and allows the condition number of the overlap matrix to enter the objective. This mirrors the original MOLOPT design: diffuse flexibility is desirable for weak interactions and response properties, but the final basis must remain numerically stable in condensed-phase simulations.

The pseudopotential optimizer constructs both all-electron and pseudopotential atomic references. Occupations are partitioned into core and valence contributions; semicore choices

* tkuehne@cp2k.org

are therefore explicit in the input and can be changed when the verification workflow reveals poor transferability. The optimizer evaluates the all-electron reference with the selected scalar-relativistic Hamiltonian, constructs the corresponding pseudopotential atom, and fits the compact GTH parameters with a derivative-free Powell algorithm. The fitted target is therefore not an isolated numerical object but a scalar-relativistic atomic mapping from an all-electron reference to the analytic separable GTH form. This machinery is what allows the UZH protocol to improve the pseudopotential component once the SIRIUS-GTH-UZH versus SIRIUS-FP-LAPW comparison identifies a pseudopotential-limited element.

B. SIRIUS calculations

SIRIUS is used in two complementary modes [9–11]. In pseudopotential plane-wave mode, SIRIUS-GTH-UZH uses the same GTH-UZH pseudopotentials as CP2K but removes the localized Gaussian-basis approximation. In all-electron full-potential linearized augmented-plane-wave (FP-LAPW) mode, SIRIUS provides the reference equation of state. The CP2K-GTH-UZH versus SIRIUS-GTH-UZH comparison isolates the CP2K Gaussian-basis error, while the two SIRIUS modes isolate the pseudopotential error.

In the FP-LAPW calculation the unit cell is divided into non-overlapping muffin-tin spheres and an interstitial region. Radial functions multiplied by spherical harmonics describe the wave functions inside the spheres, while plane waves describe the interstitial region; local orbitals are used to improve the description of semicore and high-lying states. Because the method is full-potential, the charge density and Kohn-Sham potential are represented without imposing a muffin-tin shape approximation. This all-electron reference is therefore well suited for the present purpose: any difference between SIRIUS-GTH-UZH and SIRIUS-FP-LAPW primarily measures the frozen-core and pseudization error of the GTH-UZH potential, including the quality of semicore and scalar-relativistic transferability.

For heavy elements the relativistic settings are part of the definition of the reference. The SIRIUS full-potential species files contain the radial grid, linearized augmented-plane-wave (LAPW)/augmented-plane-wave plus local-orbital (APW+lo) basis information, core-state treatment, and valence-relativity choice. The FP-LAPW reference calculations use the infinite-order regular approximation (IORA) for valence scalar relativity [12]; this IORA setting is recorded in the full-potential species definition and kept fixed throughout the equation-of-state comparison. The CP2K interface exposes the electronic-structure mode as either pseudopotential or full-potential LAPW+lo, and the full-potential setup distinguishes core relativity from valence relativity. In the UZH protocol we compare scalar-relativistic pseudopotentials to scalar-relativistic all-electron references; explicit spin-orbit splittings are outside the present benchmark. This convention keeps the error decomposition aligned with the scalar-relativistic GTH/HGH pseudopotentials used by CP2K/Quickstep.

II. MOLECULAR CALIBRATION

The molecular calibration assesses the revised UZH-MOLOPT basis sequence before the solid-state verification step. The benchmark contains small molecules covering main-group bonds, polar bonds, non-linear and dihedral angles, dipole moments, polarizabilities, and vibrational frequencies. The underlying CP2K molecular test calculations are provided in the MolTest Zenodo archive, which contains calculations with revised CP2K basis sets and pseudopotentials and CP2K all-electron basis sets for PBE, PBE0, and TPSS [13]. Figures 1, 2, and 3 provide the angular and dipole-moment statistics that complement the main-text bond-length and polarizability analysis. Figures 4 and 5 summarize two additional molecular checks from the thesis-level calibration data: the bond-class dependence of the structural errors and the consistency of Gaussian 16 and CP2K all-electron reference calculations. The role of this stage is to ensure chemically balanced transferability and numerical stability before the basis is used in periodic calculations.

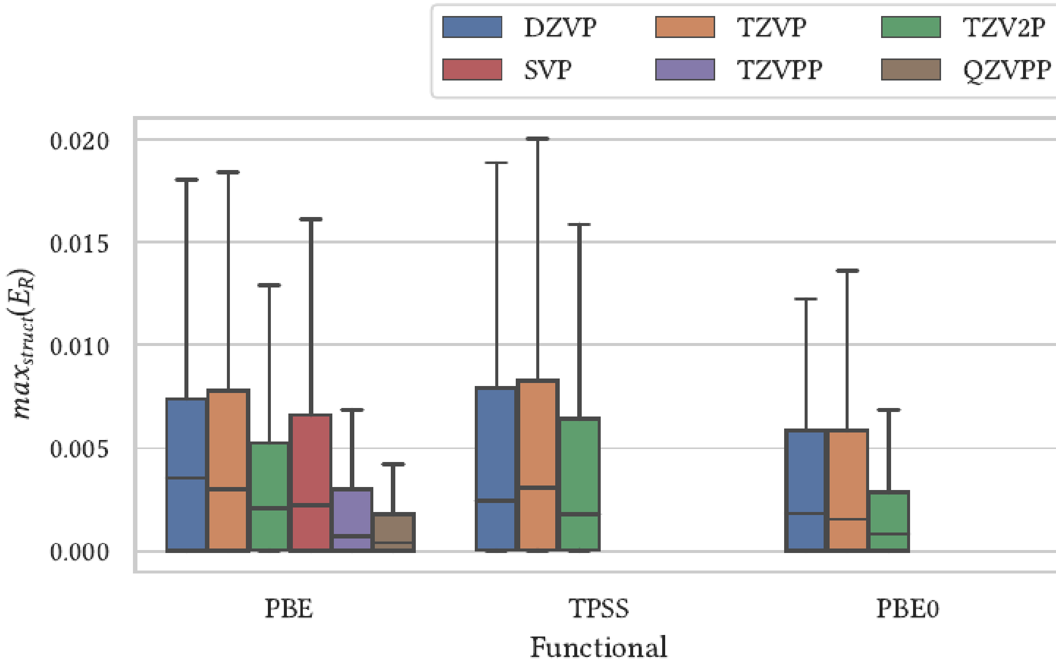


FIG. 1. Distribution of relative errors for nonlinear angles in the molecular calibration set. Angular observables are more sensitive than bond lengths to automated atom matching and near-symmetric geometries, but the systematic improvement from DZVP to TZV2P remains visible.

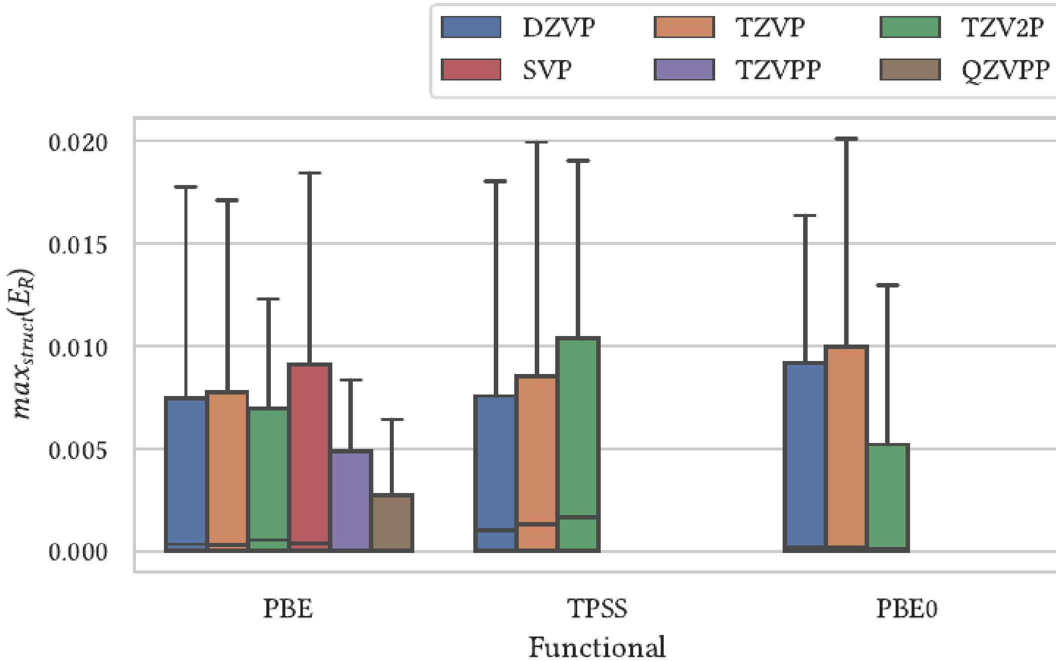


FIG. 2. Distribution of relative errors for dihedral angles in the molecular calibration set. The broader tails reflect the stronger sensitivity of torsional coordinates to small structural and matching differences, while the basis-set hierarchy remains consistent with the bond-length and nonlinear-angle tests.

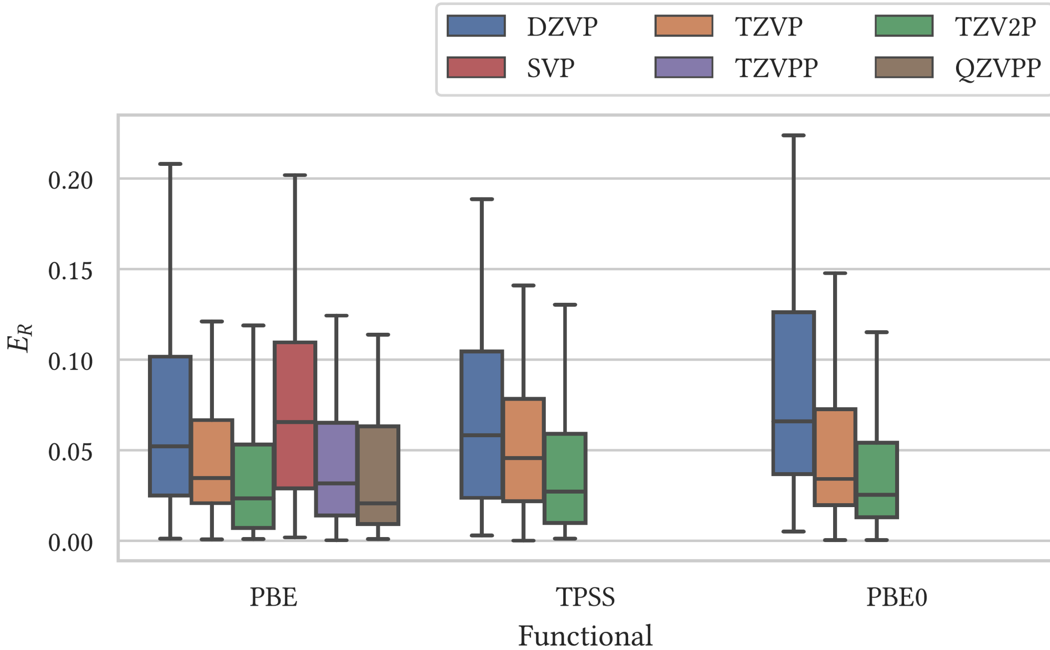


FIG. 3. Distribution of relative errors for molecular dipole moments, adapted from Fig. 4.11 of Ref. [4]. The statistics include converged structures with non-vanishing dipole moments and show that response-sensitive density properties retain the same systematic basis-set improvement observed for structural observables.

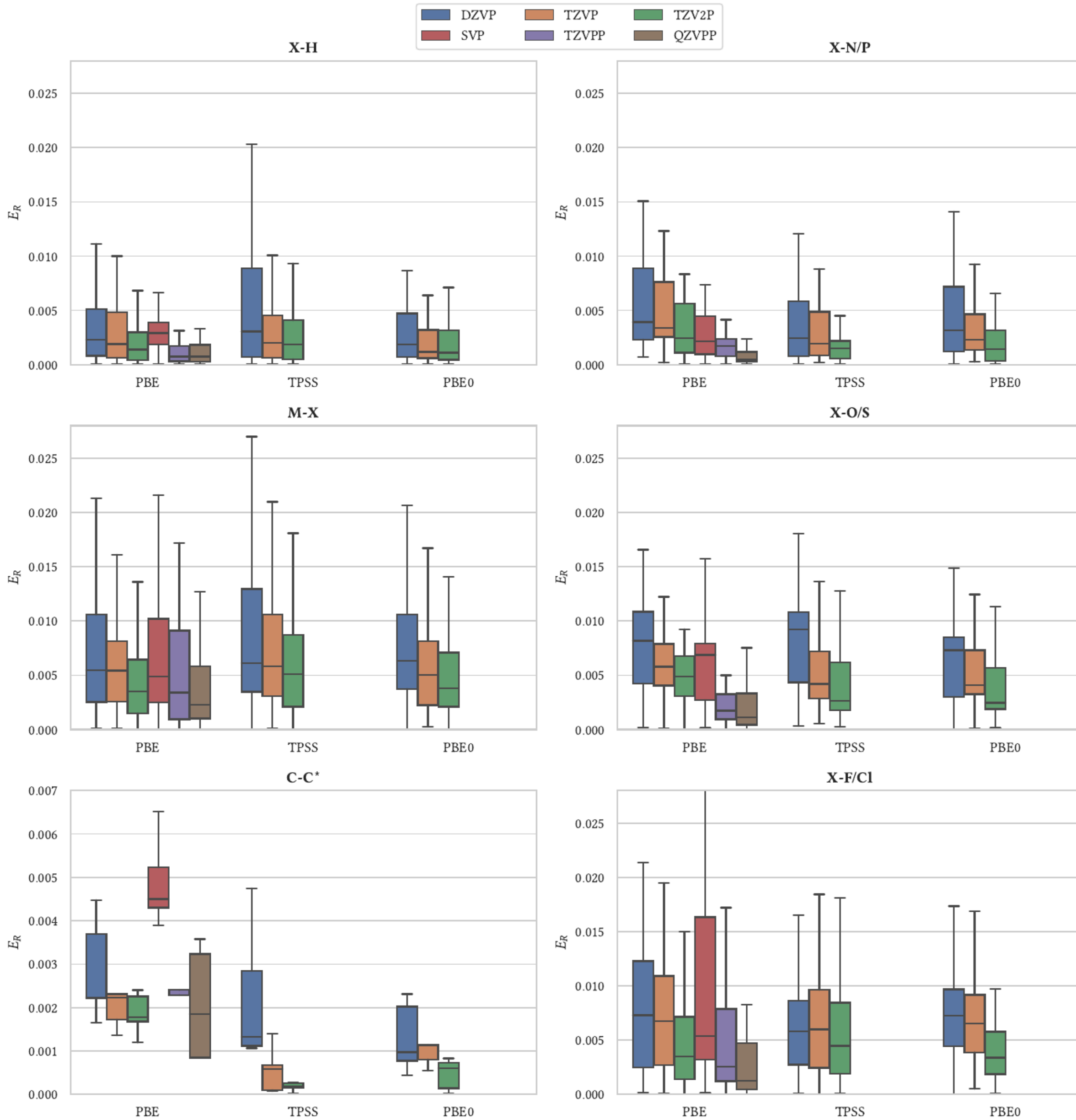


FIG. 4. Bond-class-resolved distribution of relative bond-length errors across the molecular calibration set, adapted from Fig. 4.13 of Ref. [4]. Outliers are hidden and the C–C panel uses a smaller vertical scale. The grouping shows that the UZH-MOLOPT basis sequence improves systematically across chemically distinct bond classes before periodic verification.

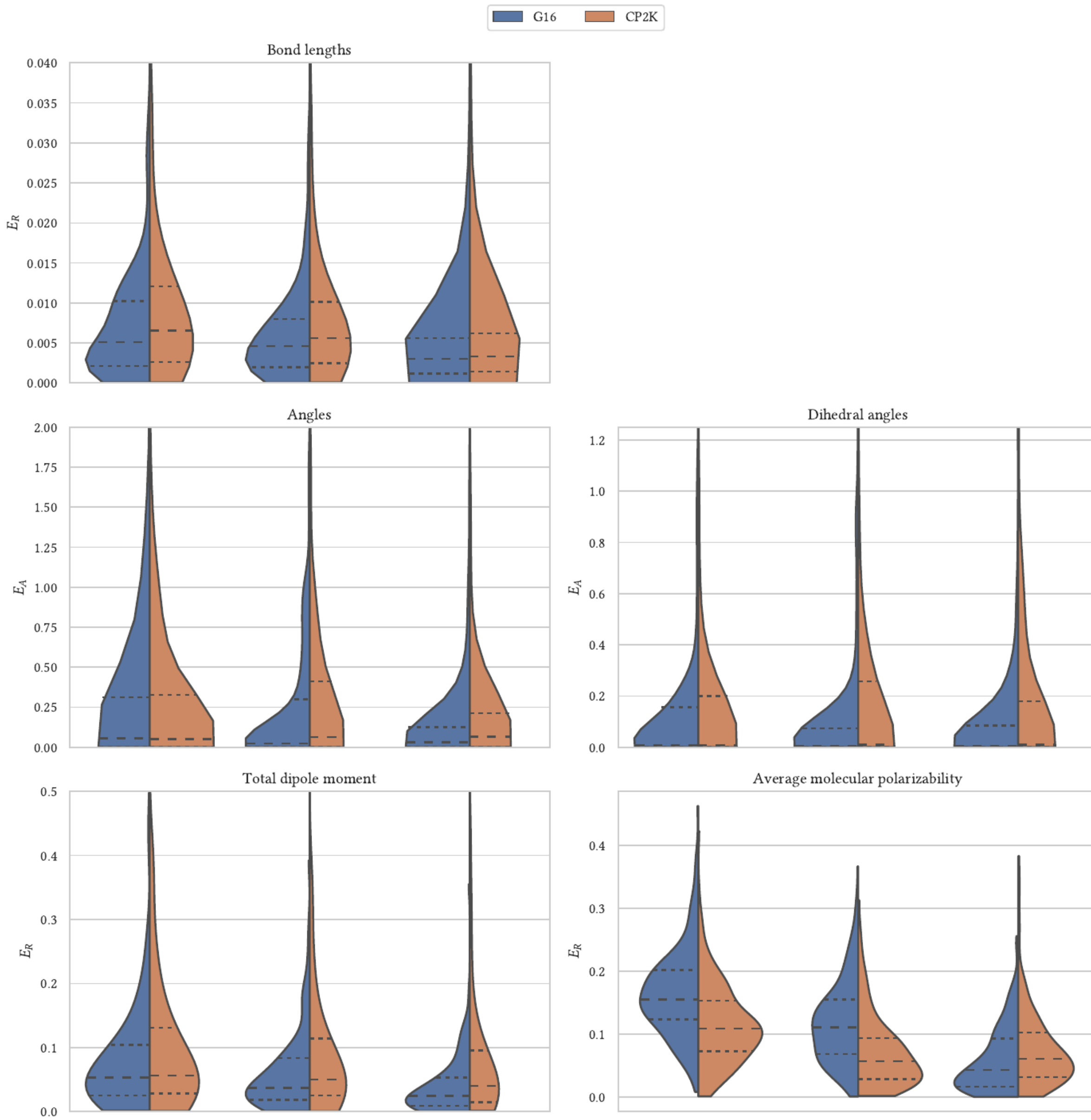
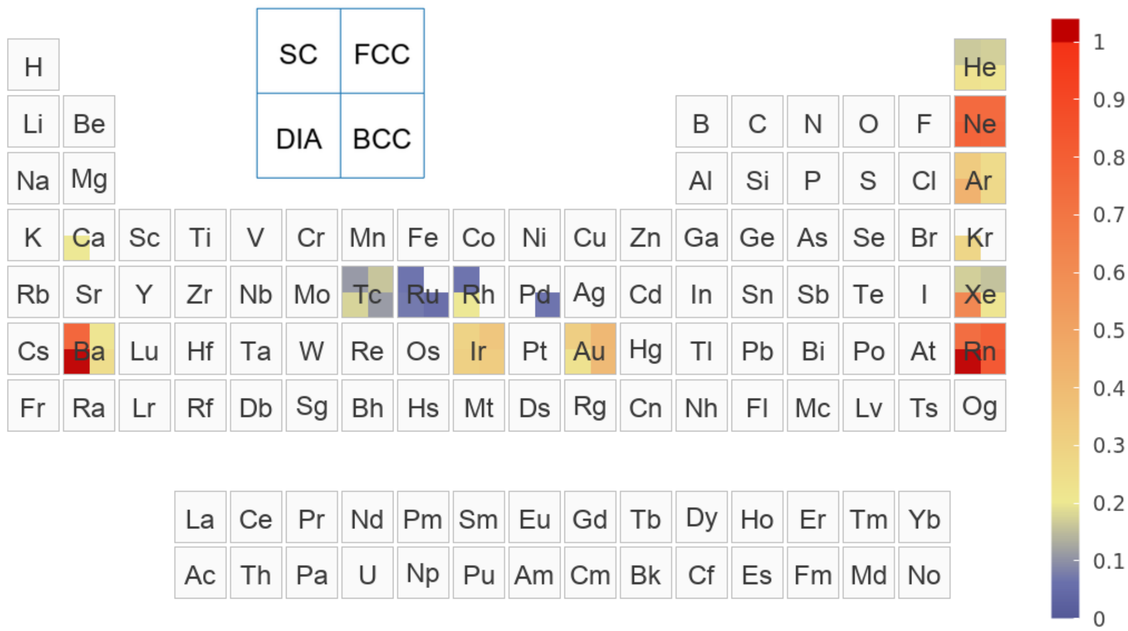


FIG. 5. Comparison of molecular calibration errors obtained with Gaussian 16/def2-QZVP and CP2K/GAPW-QZVPP all-electron references, adapted from Fig. 4.16 of Ref. [4]. Relative errors are shown for bond lengths, dipole moments, and average molecular polarizabilities, while absolute errors are shown for angles and dihedral angles. The close agreement supports the use of the CP2K all-electron molecular reference in the UZH calibration loop.

III. ADDITIONAL IMPROVED-SETTING DIAGNOSTICS



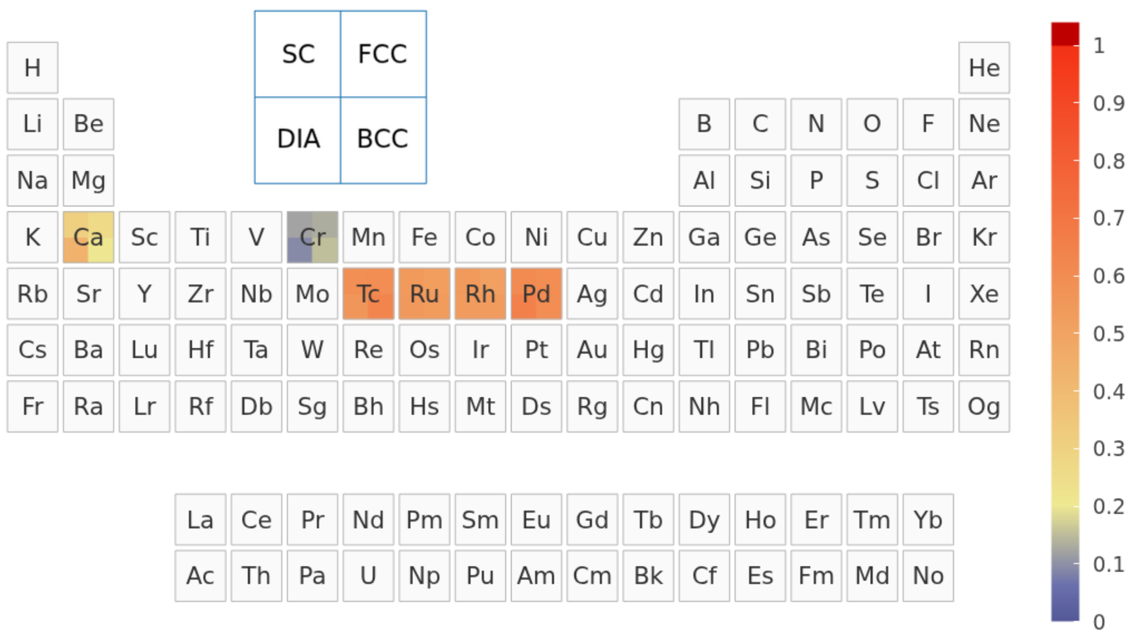


FIG. 6. Residual diagnostic for the improved settings. Top: residual CP2K Gaussian-basis contribution, constructed as the positive difference between the final CP2K-GTH-UZH-new total-error map and the residual pseudopotential map. Bottom: residual pseudopotential error for the improved SIRIUS-GTH-UZH-new settings relative to the SIRIUS-FP-LAPW reference.

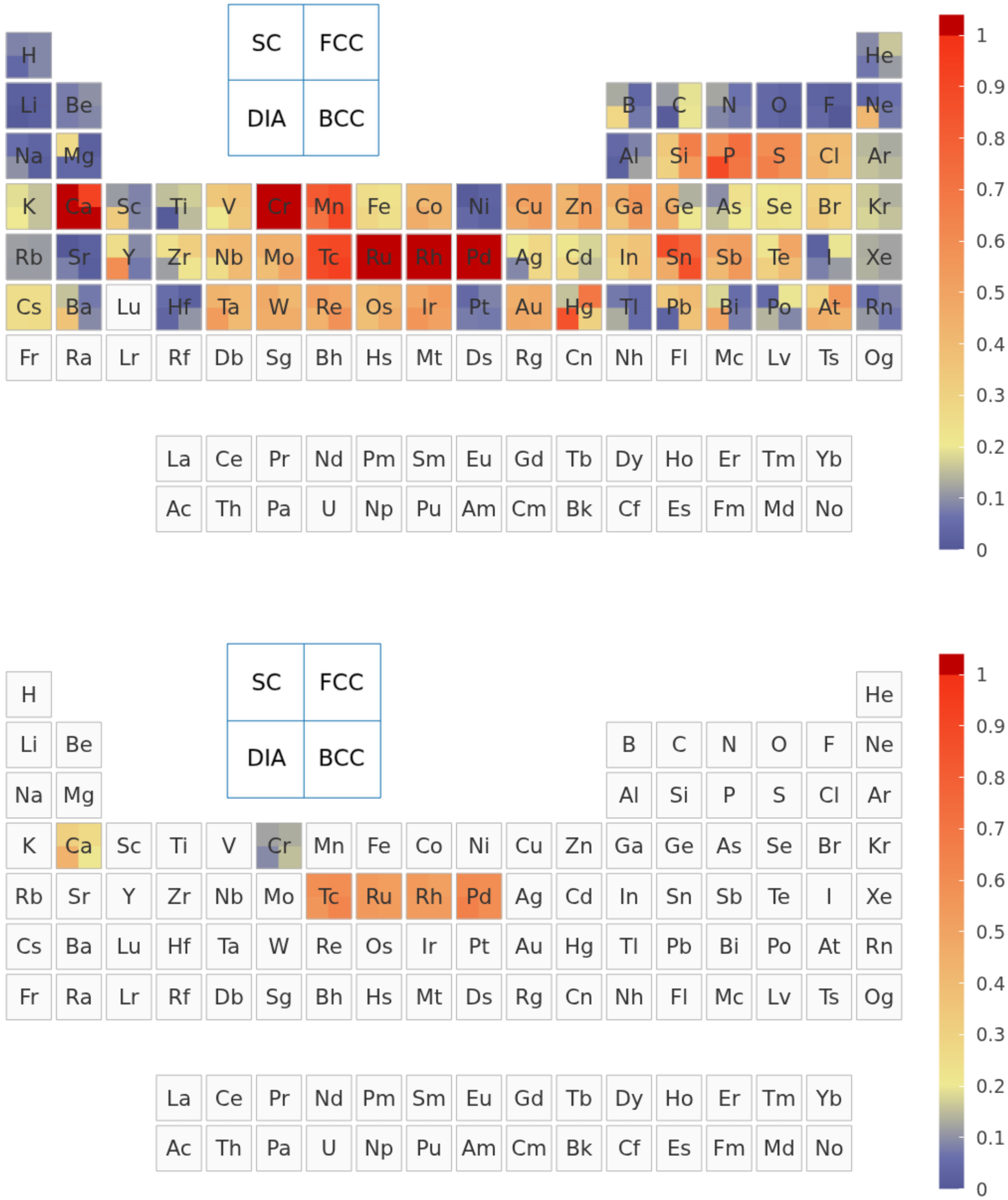


FIG. 7. Pseudopotential contribution before and after applying the UZH protocol. Top: original SIRIUS-GTH-UZH settings relative to SIRIUS-FP-LAPW. Bottom: improved SIRIUS-GTH-UZH-new settings relative to SIRIUS-FP-LAPW. Because both panels use SIRIUS on the pseudopotential side, the CP2K Gaussian basis is absent and the comparison isolates the pseudopotential transferability error.

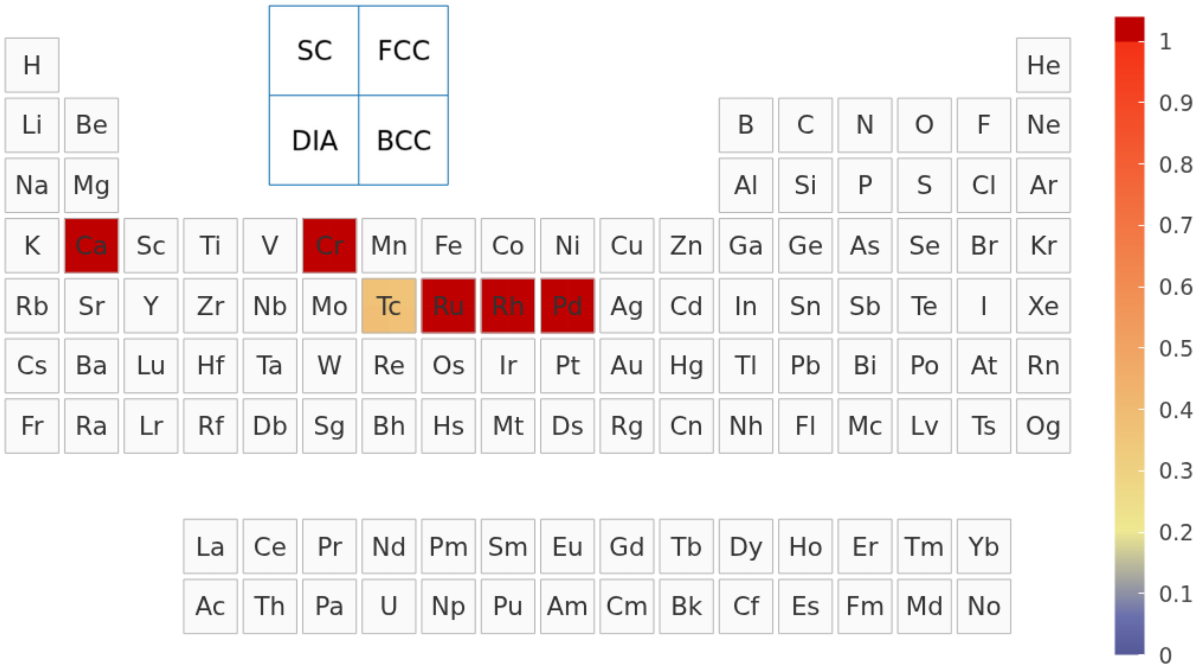


FIG. 8. Direct comparison of the improved and original SIRIUS-GTH-UZH pseudopotential settings. This diagnostic shows where the pseudopotential update changes the equation-of-state curves and complements the component-update, residual-component, and total-error maps in the main text.

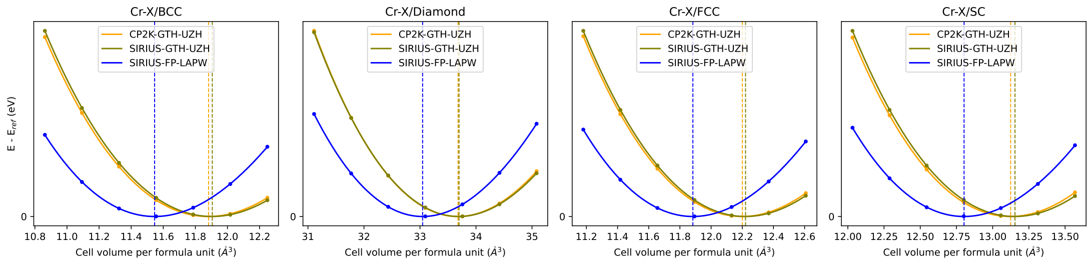


FIG. 9. Representative equation-of-state curves for Cr. The behavior is characteristic of a pseudopotential-limited case in the original settings.

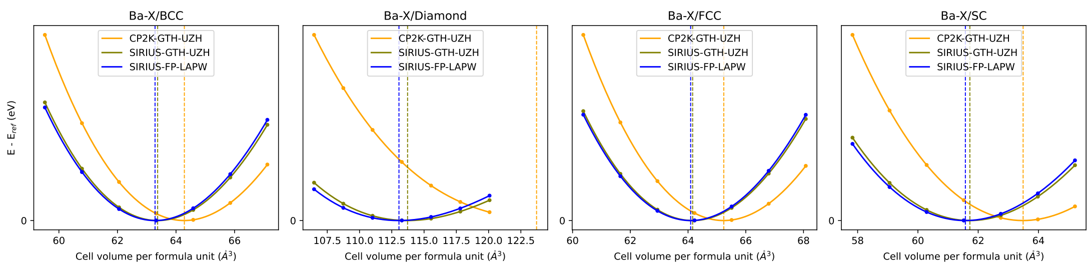


FIG. 10. Representative equation-of-state curves for Ba. The comparison highlights a case with a substantial Gaussian-basis contribution in the original CP2K-GTH-UZH settings.

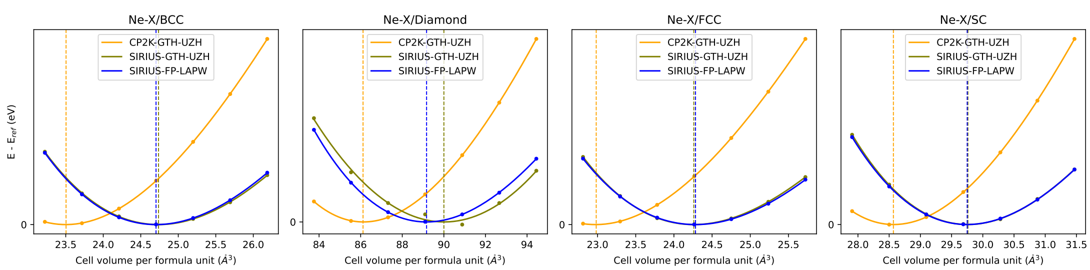


FIG. 11. Representative equation-of-state curves for Ne. Noble gases are sensitive tests of diffuse basis flexibility because their equation-of-state curves are shallow.

IV. REPRESENTATIVE EQUATION-OF-STATE CURVES

The periodic-table heat maps in the main text are compact summaries of many equation-of-state calculations. Figures 9, 10, 11, 12, and 13 show representative raw equation-of-state comparisons for elements that illustrate different error regimes.

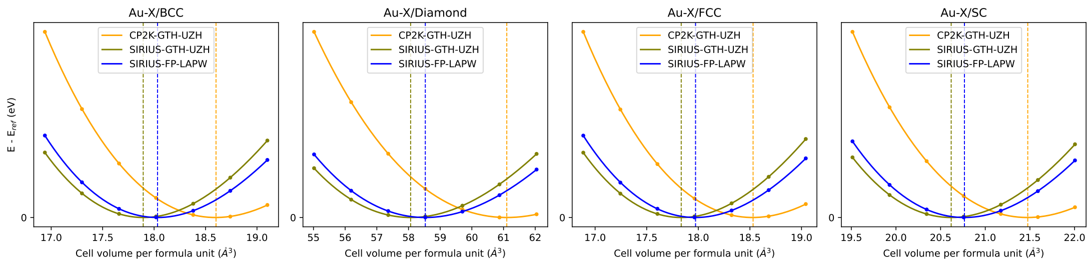


FIG. 12. Representative equation-of-state curves for Au, illustrating the behavior of a heavy element where basis completeness and relativistic pseudopotential transferability must both be monitored.

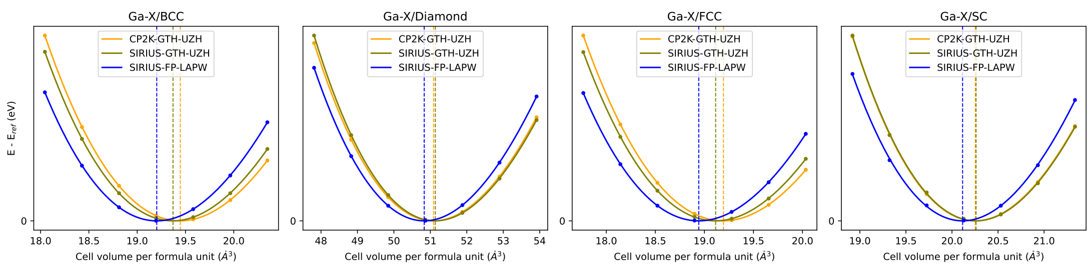


FIG. 13. Representative equation-of-state curves for Ga.

V. RELEASE REPOSITORY ORGANIZATION

The release repository for the UZH protocol is hosted in the DCM-Uni-Paderborn GitHub organization under the repository name UZH-protocol. It is organized to contain:

1. CP2K-GTH-UZH AiiDA workflow inputs and parsed results.

2. SIRIUS-GTH-UZH AiiDA workflow inputs and parsed results.

3. SIRIUS-FP-LAPW all-electron workflow inputs and parsed results.

4. Optimized UZH-MOLOPT basis files and GTH-UZH pseudopotential files.

5. Scripts used to evaluate the ε metric and generate the periodic-table maps.

6. A manifest documenting code versions, basis-file hashes, pseudopotential hashes, and workflow protocol names.

[1] J. VandeVondele and J. Hutter, Gaussian basis sets for accurate calculations on molecular systems in gas and condensed phases, The Journal of Chemical Physics **127**, 114105 (2007).

[2] S. Goedecker, M. Teter, and J. Hutter, Separable dual-space gaussian pseudopotentials, Physical Review B **54**, 1703 (1996).

[3] C. Hartwigsen, S. Goedecker, and J. Hutter, Relativistic separable dual-space gaussian pseudopotentials from h to rn, Physical Review B **58**, 3641 (1998).

[4] T. M. A. Müller, *From Benchmarking to Periodic Fock Exchange in the Auxiliary Density Matrix Method via k-Point Sampling with Gaussian Basis Sets*, Ph.D. thesis, University of Zurich (2024).

[5] T. D. Kühne, M. Iannuzzi, M. Del Ben, V. V. Rybkin, P. Seewald, F. Stein, T. Laino, R. Z. Khaliullin, O. Schütt, F. Schiffmann, D. Golze, J. Wilhelm, S. Chulkov, M. H. Bani-Hashemian, V. Weber, U. Borstnik, M. Taillefumier, A. S. Jakobovits, A. Lazzaro, H. Pabst, T. Müller, R. Schade, M. Guidon, S. Andermatt, N. Holmberg, G. K. Schenter, A. Hehn, A. Bussy, F. Belleflamme, G. Tabacchi, A. Glöß, M. Lass, I. Bethune, C. J. Mundy, C. Plessl, M. Watkins, J. VandeVondele, M. Krack, and J. Hutter, Cp2k: An electronic structure and molecular dynamics software package – quickstep: Efficient and accurate electronic structure calculations, The Journal of Chemical Physics **152**, 194103 (2020).

[6] M. Iannuzzi, J. Wilhelm, F. Stein, A. Bussy, H. Elgabarty, D. Golze, A.-S. Hehn, M. Graml, S. Marek, B. S. Gökmen, C. Schran, H. Forbert, R. Z. Khaliullin, A. Kozhevnikov, M. Taillefumier, R. Meli, V. V. Rybkin, M. Brehm, R. Schade, O. Schütt, J. V. Pototschnig, H. Mirhosseini, A. Knüpfer, D. Marx, M. Krack, J. Hutter, and T. D. Kühne, The cp2k program package made simple, The Journal of Physical Chemistry B **130**, 1237 (2026).

[7] S. P. Huber, E. Bosoni, M. Bercx, J. Bröder, A. Degomme, V. Dikan, K. Eimre, E. Flage-Larsen, A. Garcia, L. Genovese, D. Gresch, C. Johnston, G. Petretto, S. Poncé, G.-M. Rignanese, C. J. Sewell, B. Smit, V. Tseplyaev, M. Uhrin, D. Wortmann, A. V. Yakutovich, A. Zadoks, P. Zarabadi-Poor, B. Zhu, N. Marzari, and G. Pizzi, Common workflows for computing material properties using different quantum engines, npj Computational Materials **7**, 136 (2021).

[8] E. Bosoni, L. Beal, M. Bercx, P. Blaha, S. Blügel, J. Bröder, M. Callsen, S. Cottenier, A. De-

gomme, V. Dikan, K. Eimre, E. Flage-Larsen, M. Fornari, A. Garcia, L. Genovese, M. Giantomassi, S. P. Huber, H. Janssen, G. Kastlunger, M. Krack, G. Kresse, T. D. Kühne, K. Lejaeghere, G. K. H. Madsen, M. Marsman, N. Marzari, G. Michalicek, H. Mirhosseini, T. M. A. Müller, G. Petretto, C. J. Pickard, S. Poncé, G.-M. Rignanese, O. Rubel, T. Ruh, M. Sluydts, D. E. P. Vanpoucke, S. Vijay, M. Wolloch, D. Wortmann, A. V. Yakutovich, J. Yu, A. Zadoks, B. Zhu, and G. Pizzi, How to verify the precision of density-functional-theory implementations via reproducible and universal workflows, Nature Reviews Physics **6**, 45 (2024).

[9] L. Zhang, A. Kozhevnikov, T. Schulthess, H.-P. Cheng, and S. B. Trickey, Performance enhancement of apw+lo calculations by simplest separation of concerns, Computation **10**, 43 (2022).

[10] L. Zhang, A. Kozhevnikov, T. Schulthess, S. B. Trickey, and H.-P. Cheng, All-electron apw+lo calculation of magnetic molecules with the sirius domain-specific package, The Journal of Chemical Physics **158**, 234801 (2023).

[11] SIRIUS developers, SIRIUS: Domain specific library for electronic structure calculations (2026), documentation and source repository, `https://electronic-structure.github.io/SIRIUS-doc/` and `https://github.com/electronic-structure/SIRIUS`.

[12] K. G. Dyall and E. van Lenthe, Relativistic regular approximations revisited: An infinite-order relativistic approximation, The Journal of Chemical Physics **111**, 1366 (1999).

[13] J. Hutter and T. M. A. Müller, CP2K calculations for structures of the small molecule database with the revised MOLOPT basis sets and pseudopotentials, Zenodo (2023), version 1.0, dataset.